5.3 Geometric interpretation

5.3.1 Weak and strong duality via set of values

We can give a simple geometric interpretation of the dual function in terms of the set

$$\mathcal{G} = \{(f_1(x), \ldots, f_m(x), h_1(x), \ldots, h_p(x), f_0(x)) \in \mathbf{R}^m \times \mathbf{R}^p \times \mathbf{R} \mid x \in \mathcal{D}\}, \quad (5.36)$$

which is the set of values taken on by the constraint and objective functions. The optimal value $p^\star$ of (5.1) is easily expressed in terms of $\mathcal{G}$ as

$$p^\star = \inf\{t \mid (u, v, t) \in \mathcal{G},\ u \preceq 0,\ v = 0\}.$$

To evaluate the dual function at (λ, ν), we minimize the affine function

$$(\lambda, \nu, 1)^T(u, v, t) = \sum_{i=1}^m \lambda_i u_i + \sum_{i=1}^p \nu_i v_i + t$$

over $(u, v, t) \in \mathcal{G}$, *i.e.*, we have

$$g(\lambda, \nu) = \inf\{(\lambda, \nu, 1)^T(u, v, t) \mid (u, v, t) \in \mathcal{G}\}.$$

In particular, we see that if the infimum is finite, then the inequality

$$(\lambda, \nu, 1)^T(u, v, t) \geq g(\lambda, \nu)$$

defines a supporting hyperplane to $\mathcal{G}$. This is sometimes referred to as a *nonvertical* supporting hyperplane, because the last component of the normal vector is nonzero.

Now suppose $\lambda \succeq 0$. Then, obviously, $t \geq (\lambda, \nu, 1)^T(u, v, t)$ if $u \preceq 0$ and $v = 0$. Therefore

$$\begin{array}{lcl} p^\star & = & \inf\{t \mid (u, v, t) \in \mathcal{G},\ u \preceq 0,\ v = 0\} \\ & \geq & \inf\{(\lambda, \nu, 1)^T(u, v, t) \mid (u, v, t) \in \mathcal{G},\ u \preceq 0,\ v = 0\} \\ & \geq & \inf\{(\lambda, \nu, 1)^T(u, v, t) \mid (u, v, t) \in \mathcal{G}\} \\ & = & g(\lambda, \nu), \end{array}$$

i.e., we have weak duality. This interpretation is illustrated in figures 5.3 and 5.4, for a simple problem with one inequality constraint.

Epigraph variation

In this section we describe a variation on the geometric interpretation of duality in terms of $\mathcal{G}$, which explains why strong duality obtains for (most) convex problems. We define the set $\mathcal{A} \subseteq \mathbf{R}^m \times \mathbf{R}^p \times \mathbf{R}$ as

$$\mathcal{A} = \mathcal{G} + \left(\mathbf{R}_+^m \times \{0\} \times \mathbf{R}_+\right), \quad (5.37)$$

or, more explicitly,

$$\begin{array}{l} \mathcal{A} = \{(u, v, t) \mid \exists x \in \mathcal{D},\ f_i(x) \leq u_i,\ i = 1, \ldots, m, \\ \qquad h_i(x) = v_i,\ i = 1, \ldots, p,\ f_0(x) \leq t\}, \end{array}$$

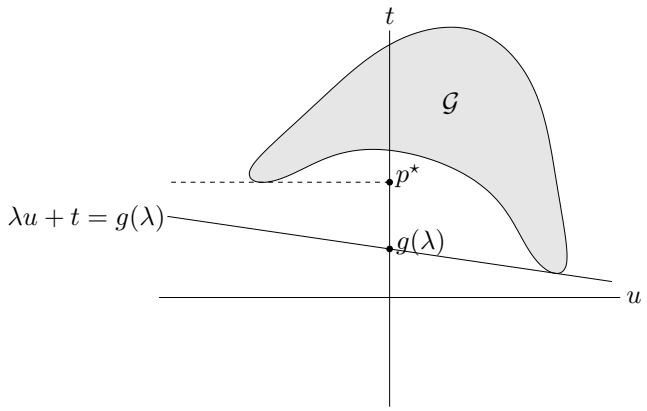


Figure 5.3 Geometric interpretation of dual function and lower bound $g(\lambda) \leq p^\star$, for a problem with one (inequality) constraint. Given λ, we minimize $(\lambda, 1)^T(u, t)$ over $\mathcal{G} = \{(f_1(x), f_0(x)) \mid x \in \mathcal{D}\}$. This yields a supporting hyperplane with slope $-\lambda$. The intersection of this hyperplane with the $u = 0$ axis gives $g(\lambda)$.

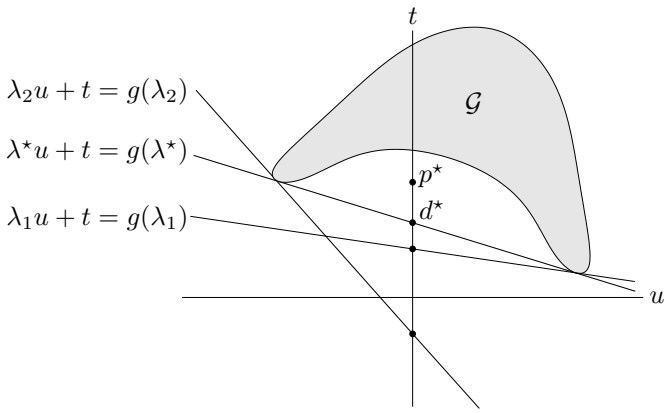


Figure 5.4 Supporting hyperplanes corresponding to three dual feasible values of λ, including the optimum $\lambda^\star$. Strong duality does not hold; the optimal duality gap $p^\star - d^\star$ is positive.

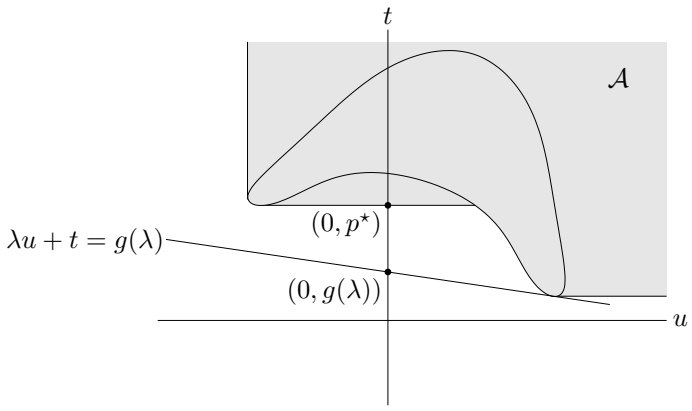


Figure 5.5 Geometric interpretation of dual function and lower bound $g(\lambda) \leq p^\star$, for a problem with one (inequality) constraint. Given λ, we minimize $(\lambda, 1)^T(u, t)$ over $\mathcal{A} = \{(u, t) \mid \exists x \in \mathcal{D},\ f_0(x) \leq t,\ f_1(x) \leq u\}$. This yields a supporting hyperplane with slope $-\lambda$. The intersection of this hyperplane with the $u = 0$ axis gives $g(\lambda)$.

We can think of $\mathcal{A}$ as a sort of epigraph form of $\mathcal{G}$, since $\mathcal{A}$ includes all the points in $\mathcal{G}$, as well as points that are 'worse', *i.e.*, those with larger objective or inequality constraint function values.

We can express the optimal value in terms of $\mathcal{A}$ as

$$p^\star = \inf\{t \mid (0, 0, t) \in \mathcal{A}\}.$$

To evaluate the dual function at a point (λ, ν) with $\lambda \succeq 0$, we can minimize the affine function $(\lambda, \nu, 1)^T(u, v, t)$ over $\mathcal{A}$: If $\lambda \succeq 0$, then

$$g(\lambda, \nu) = \inf\{(\lambda, \nu, 1)^T(u, v, t) \mid (u, v, t) \in \mathcal{A}\}.$$

If the infimum is finite, then

$$(\lambda, \nu, 1)^T(u, v, t) \geq g(\lambda, \nu)$$

defines a nonvertical supporting hyperplane to $\mathcal{A}$.

In particular, since $(0, 0, p^\star) \in \mathbf{bd}\, \mathcal{A}$, we have

$$p^\star = (\lambda, \nu, 1)^T(0, 0, p^\star) \geq g(\lambda, \nu), \tag{5.38}$$

the weak duality lower bound. Strong duality holds if and only if we have equality in (5.38) for some dual feasible (λ, ν), *i.e.*, there exists a nonvertical supporting hyperplane to $\mathcal{A}$ at its boundary point $(0, 0, p^\star)$.

This second interpretation is illustrated in figure 5.5.

5.3.2 Proof of strong duality under constraint qualification

In this section we prove that Slater's constraint qualification guarantees strong duality (and that the dual optimum is attained) for a convex problem. We consider

the primal problem (5.25), with $f_0, \ldots, f_m$ convex, and assume Slater's condition holds: There exists $\tilde{x} \in \mathbf{relint}\,\mathcal{D}$ with $f_i(\tilde{x}) < 0$, $i = 1, \ldots, m$, and $A\tilde{x} = b$. In order to simplify the proof, we make two additional assumptions: first that $\mathcal{D}$ has nonempty interior (hence, $\mathbf{relint}\,\mathcal{D} = \mathbf{int}\,\mathcal{D}$) and second, that $\mathbf{rank}\,A = p$. We assume that $p^\star$ is finite. (Since there is a feasible point, we can only have $p^\star = -\infty$ or $p^\star$ finite; if $p^\star = -\infty$, then $d^\star = -\infty$ by weak duality.)

The set $\mathcal{A}$ defined in (5.37) is readily shown to be convex if the underlying problem is convex. We define a second convex set $\mathcal{B}$ as

$$\mathcal{B} = \{(0, 0, s) \in \mathbf{R}^m \times \mathbf{R}^p \times \mathbf{R} \mid s < p^\star\}.$$

The sets $\mathcal{A}$ and $\mathcal{B}$ do not intersect. To see this, suppose $(u, v, t) \in \mathcal{A} \cap \mathcal{B}$. Since $(u, v, t) \in \mathcal{B}$ we have $u = 0$, $v = 0$, and $t < p^\star$. Since $(u, v, t) \in \mathcal{A}$, there exists an x with $f_i(x) \leq 0$, $i = 1, \ldots, m$, $Ax - b = 0$, and $f_0(x) \leq t < p^\star$, which is impossible since $p^\star$ is the optimal value of the primal problem.

By the separating hyperplane theorem of §2.5.1 there exists $(\tilde{\lambda}, \tilde{\nu}, \mu) \neq 0$ and α such that

$$(u, v, t) \in \mathcal{A} \implies \tilde{\lambda}^T u + \tilde{\nu}^T v + \mu t \geq \alpha, \tag{5.39}$$

and

$$(u, v, t) \in \mathcal{B} \implies \tilde{\lambda}^T u + \tilde{\nu}^T v + \mu t \leq \alpha. \tag{5.40}$$

From (5.39) we conclude that $\tilde{\lambda} \succeq 0$ and $\mu \geq 0$. (Otherwise $\tilde{\lambda}^T u + \mu t$ is unbounded below over $\mathcal{A}$, contradicting (5.39).) The condition (5.40) simply means that $\mu t \leq \alpha$ for all $t < p^\star$, and hence, $\mu p^\star \leq \alpha$. Together with (5.39) we conclude that for any $x \in \mathcal{D}$,

$$\sum_{i=1}^m \tilde{\lambda}_i f_i(x) + \tilde{\nu}^T (Ax - b) + \mu f_0(x) \geq \alpha \geq \mu p^\star. \tag{5.41}$$

Assume that $\mu > 0$. In that case we can divide (5.41) by μ to obtain

$$L(x, \tilde{\lambda}/\mu, \tilde{\nu}/\mu) \geq p^\star$$

for all $x \in \mathcal{D}$, from which it follows, by minimizing over x, that $g(\lambda, \nu) \geq p^\star$, where we define

$$\lambda = \tilde{\lambda}/\mu, \qquad \nu = \tilde{\nu}/\mu.$$

By weak duality we have $g(\lambda, \nu) \leq p^\star$, so in fact $g(\lambda, \nu) = p^\star$. This shows that strong duality holds, and that the dual optimum is attained, at least in the case when $\mu > 0$.

Now consider the case $\mu = 0$. From (5.41), we conclude that for all $x \in \mathcal{D}$,

$$\sum_{i=1}^m \tilde{\lambda}_i f_i(x) + \tilde{\nu}^T (Ax - b) \geq 0. \tag{5.42}$$

Applying this to the point $\tilde{x}$ that satisfies the Slater condition, we have

$$\sum_{i=1}^m \tilde{\lambda}_i f_i(\tilde{x}) \geq 0.$$

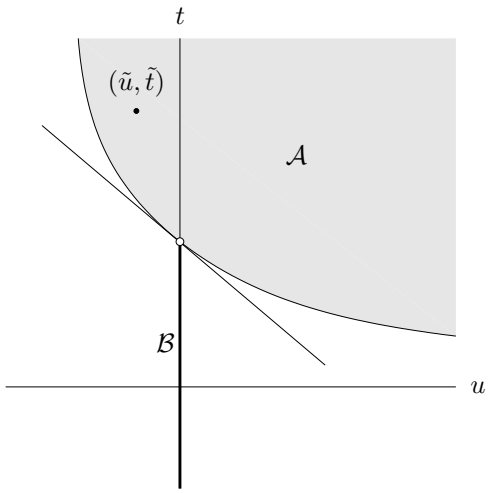


Figure 5.6 Illustration of strong duality proof, for a convex problem that satisfies Slater's constraint qualification. The set $\mathcal{A}$ is shown shaded, and the set $\mathcal{B}$ is the thick vertical line segment, not including the point $(0, p^\star)$, shown as a small open circle. The two sets are convex and do not intersect, so they can be separated by a hyperplane. Slater's constraint qualification guarantees that any separating hyperplane must be nonvertical, since it must pass to the left of the point $(\tilde{u}, \tilde{t}) = (f_1(\tilde{x}), f_0(\tilde{x}))$, where $\tilde{x}$ is strictly feasible.

Since $f_i(\tilde{x}) < 0$ and $\tilde{\lambda}_i \geq 0$, we conclude that $\tilde{\lambda} = 0$. From $(\tilde{\lambda}, \tilde{\nu}, \mu) \neq 0$ and $\tilde{\lambda} = 0$, $\mu = 0$, we conclude that $\tilde{\nu} \neq 0$. Then (5.42) implies that for all $x \in \mathcal{D}$, $\tilde{\nu}^T(Ax - b) \geq 0$. But $\tilde{x}$ satisfies $\tilde{\nu}^T(A\tilde{x} - b) = 0$, and since $\tilde{x} \in \mathbf{int}\,\mathcal{D}$, there are points in $\mathcal{D}$ with $\tilde{\nu}^T(Ax - b) < 0$ unless $A^T\tilde{\nu} = 0$. This, of course, contradicts our assumption that $\mathbf{rank}\, A = p$.

The geometric idea behind the proof is illustrated in figure 5.6, for a simple problem with one inequality constraint. The hyperplane separating $\mathcal{A}$ and $\mathcal{B}$ defines a supporting hyperplane to $\mathcal{A}$ at $(0, p^\star)$. Slater's constraint qualification is used to establish that the hyperplane must be nonvertical (*i.e.*, has a normal vector of the form $(\lambda^\star, 1)$). (For a simple example of a convex problem with one inequality constraint for which strong duality fails, see exercise 5.21.)

5.3.3 Multicriterion interpretation

There is a natural connection between Lagrange duality for a problem without equality constraints,

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m, \end{array} \tag{5.43}$$